Home Energy Audit

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ENG6079 – Thermodynamics and Energy Systems

BEng Mechanical Engineering **18/05/2021**

Abstract

A home energy audit was carried out on a 5 bedroom house recording consumption of all appliances to critically analyze consumption by appliance and against the energy bill, as well as carrying out a carbon footprint analysis, The highest consumption natural gas appliances were found to be the boiler and cooker, while for electrical the lighting and refrigeration consumed the most energy. The house had an average calculated efficiency of 80.91%. Electricity accounted for 19.17% of the total quarterly energy consumption of 8182.8kWh but cost 52.28% of the energy bill of £388.83. Recommended improvements included improving lighting and refrigeration efficiency, switching to electric heating, using an induction cooker, solar panel installation and lowering the boiler temperature to 60°C. The carbon footprint analysis showed a footprint of 1.248 Tonnes of CO_2 for the household, excluding vehicles and external factors. Electricity costs 4.8 times more than natural gas which was economically unviable. Switching to the nuclear source reduced the footprint to 0.163 Tonnes annually. For renewable energy to be adopted, energy providers had to reduce the cost of electricity.

Carbon Footprint Analysis, Efficiency, Solar Panels, Home Energy Audit, Energy Ethics.

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Main Body

Introduction

Objectives

- Estimate the itemized power consumption of a 5-bedroom house with 3 occupants, calculating the efficiency of the home using the energy bill.
- Visualise the consumption and provide insights to findings, as well as performing a carbon footprint analysis
- Demonstrate how the home energy audit can be used to make reductions, discussing surrounding issues.

Background

A lot of consumers spend time comparing energy providers to save money on their utility bills but hardly consider auditing their own home for energy consumption. By visualizing the energy consumption of the home, changes can be made to actively reduce consumption thus carbon footprint.

Rationale

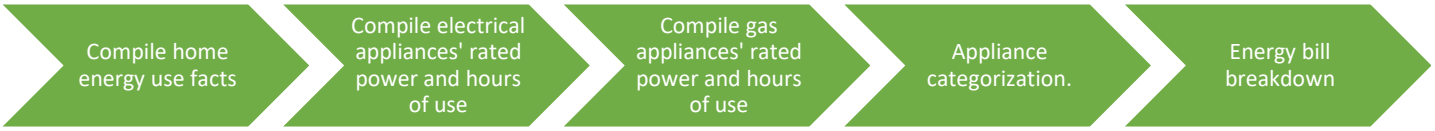
To understand how the home energy audit is beneficial for homeowners and evaluate the issues with optimizing energy management.

Literature Review

The research paper by Jawad (2018) had shown an efficient process of carrying out a home energy audit which was used as inspiration for the methodology, while the paper by Kumar et al (2015), provided inspiration for the visualisation of the data.

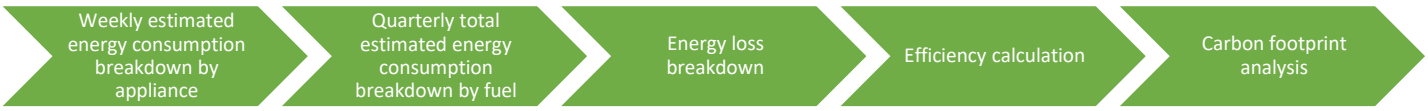
Methodology

Data Collection Methodology



Data Pre-processing Methodology

Data Visualisation Methodology



Equations used.

The following equations were used as part of the calculations for the home energy audit.

Calculation	Equation
Power	$P = I * V$
Energy	$Q = I * t$
Energy loss	$Q_{loss} = Q_{out} - Q_{in}$
Efficiency	$\eta = \frac{Q_{out}}{Q_{in}} * 100$

Assumptions made

The assumptions made during the home energy audit are outlined below.

System	Intermittent use?	Consumption assumptions
Hot water	✔	12.5% of est. gas consumption.
Refrigeration	✔ ✔	Fridge - EPC grade E, consuming 130kWh annually (Miele - K 37672 iD – Refrigeration, 2021). Freezer - EPC grade F, consuming 150kWh annually (Electrolux - Integrated freezer - LUT6NF18C, 2021).
Boiler	✔	92% efficiency Baxi Duo Tec 40 (Baxi Duo-tec Combi | Baxi, 2021).
Pipework	✔	90% efficiency, cannot inspect actual pipework (Understanding Central Heating Systems, 2021).

Cooker	✗	55.83% efficiency, average off internet. (Naveen et al., 2015)
Space heating	✓	The remainder of gas usage after accounting for cooking, hot water and losses.
Other electrical	✗	Losses assumed to be equal to reduce complexity.

The accuracy of the data collection is outlined below.

Data type	Decimal places
Calculations	3
Totals	2
Results	2

Results and Discussion

Weekly Itemised Energy Consumption Breakdown.

Electricity Analysis

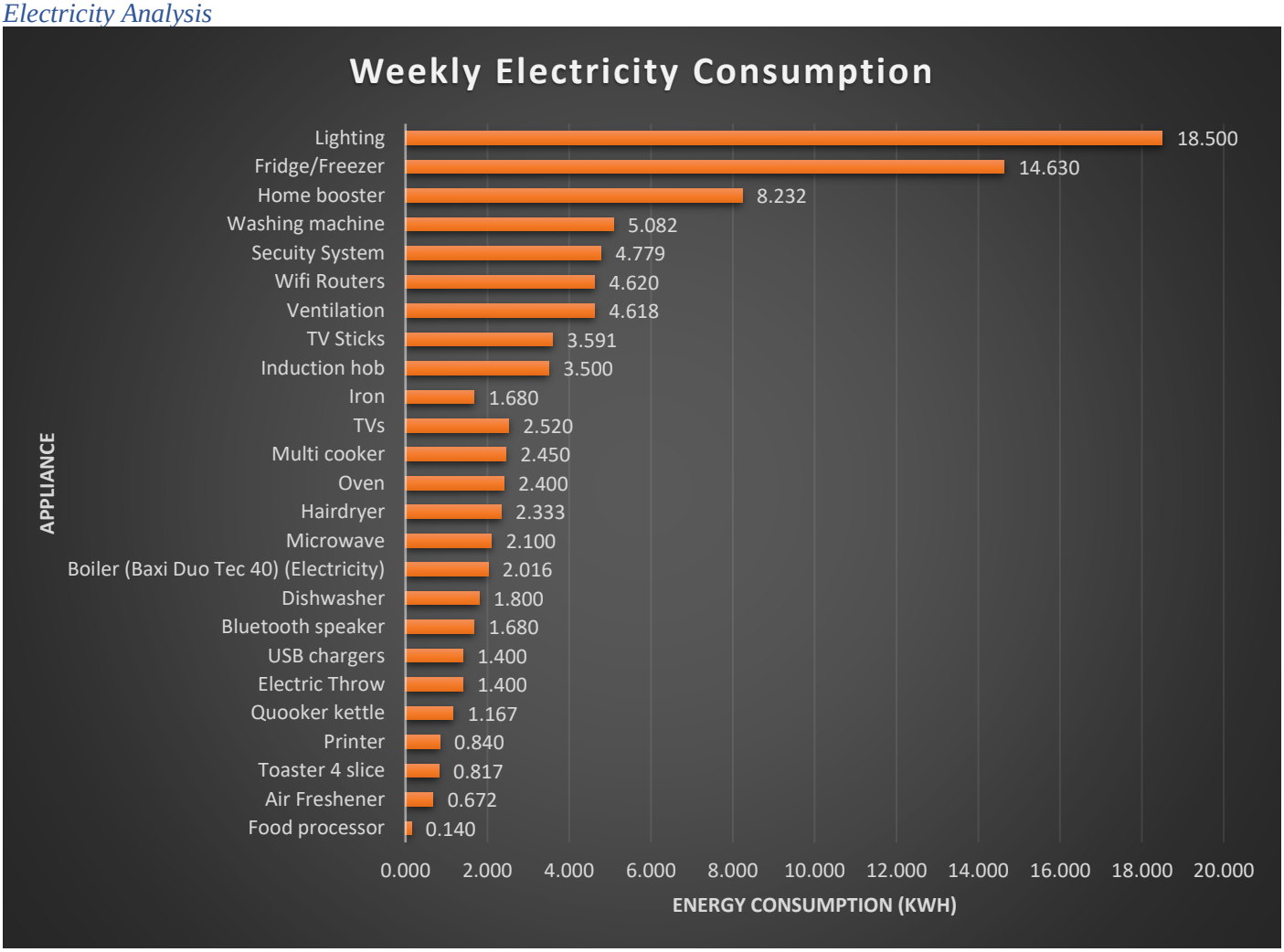


Figure 1: Weekly Itemised Electricity Consumption Breakdown

The appliances with the highest energy consumption were analysed for optimisation, as shown.

Lighting

The house had several spotlights fitted in all the rooms, with one switch and no dimmers, resulting in a high energy consumption. To reduce consumption, the following solutions were proposed:

- **Fit dimmer switches to the lighting -** Dimmer switches reduce the current available to the bulbs.
- **Use less bulbs -** Using floor or table lamps when full lighting is not necessary.
- **Fit PIR sensors to the lighting -** PIR sensors will only switch the lights on when movement is detected.

Refrigeration

Appliance	Energy Rating	Annual Energy Consumption
Fridge	E	155.86
Freezer	F	292

Figure 3: Refrigeration Energy Consumption

As reflected by figure 1 and 3, the low efficiency refrigeration appliances consumed the 2nd most electricity, of the appliances. To reduce consumption, the following solutions were proposed:

- **Purchase more efficient refrigeration -** Modern refrigeration appliances have efficient compressors and improved insulation, hence more efficient refrigeration cycles.
- **Close the door -** Closing the door minimises the gain in temperature, shortening the refrigeration cycle.

Home booster

The home booster was in the detached garage. It spent the majority the day recirculating the central heating system, with occasional hot water use. To reduce consumption, the following solutions were proposed:

- **Move the booster into the house -** The house is insulated, while the garage is not, therefore there will be less heat loss.
- **Switch to electric heating -** Eliminating the central heating system, the booster would only be used for hot water.

More tips

TV sticks can use just as much energy as an induction hob, to save energy:

- Switch appliances off fully when possible.
- Switching off Wi-Fi routers at night, reduces energy consumption by a third.

Gas Analysis

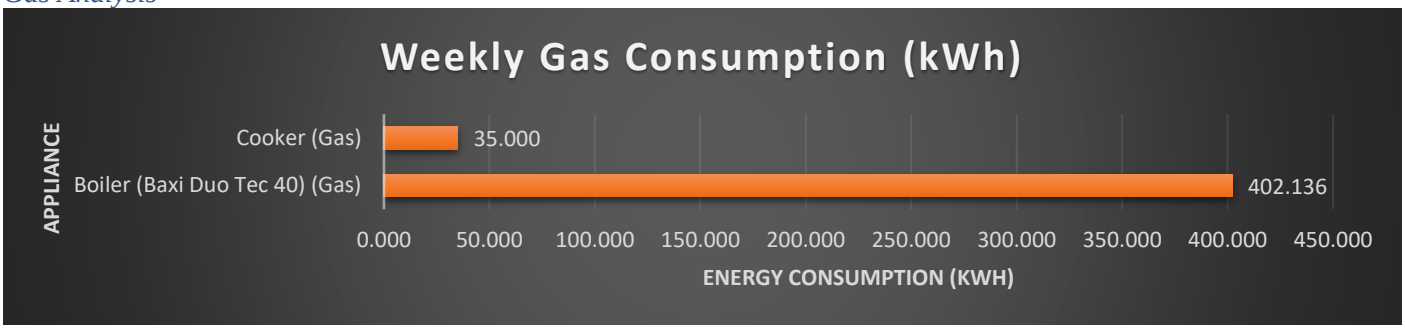


Figure 2: Weekly Itemised Gas Consumption Breakdown

Cooker

It was found that gas cookers were only 53.83% efficient (Open University, 2021) with the rest of the energy lost as heat and light. To reduce consumption, the following solutions were proposed:

- **Use the induction cooker -** Induction cookers have minimal heat loss as the pan is heated as a result of magnetic induction, providing more useful energy.

Boiler

The boiler loses most of its energy heating the water used for central heating. To reduce consumption, the following solutions were proposed:

- **Switch to electric heating -** The majority of the boiler's runtime was for the central heating system. In contrast, electric heaters heat rooms directly, with no energy losses.
- **Lower the water temperature on the boiler -** The temperature can be lowered to a limit of 60°C due to a risk of "Legionella infection" (Borella et al., 2004).

Weekly Categorised Energy Consumption Breakdown

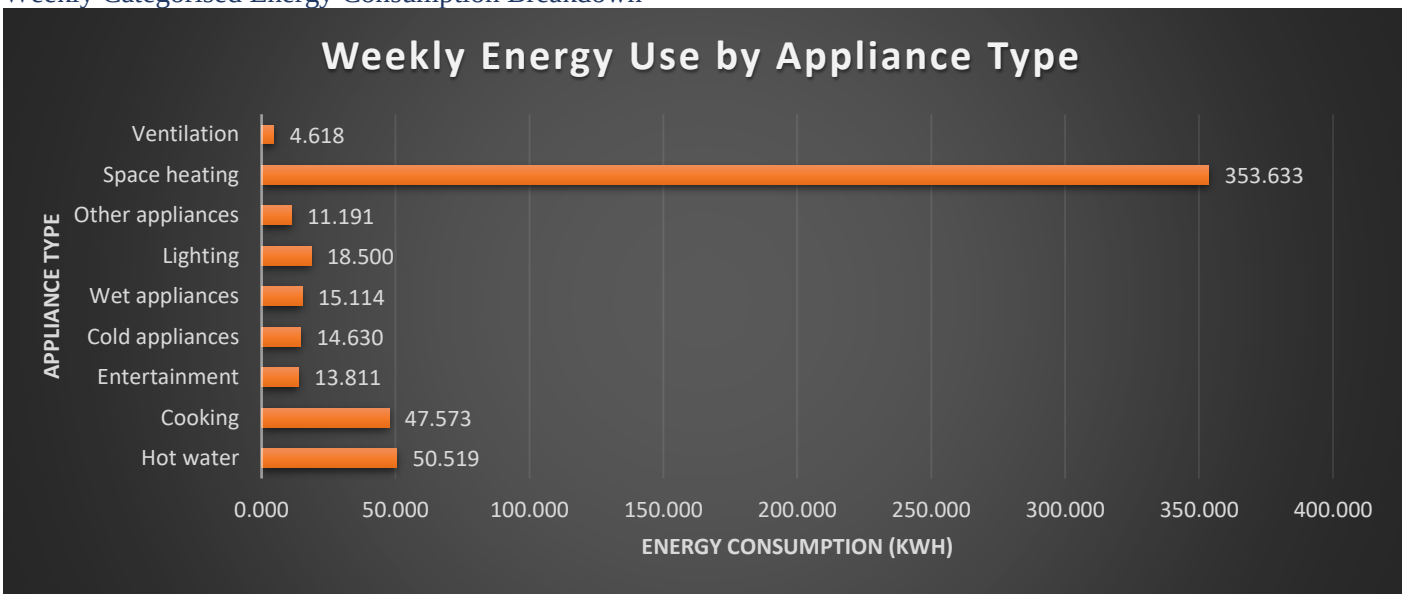


Figure 3: Categorical Breakdown of Weekly Energy Consumption.

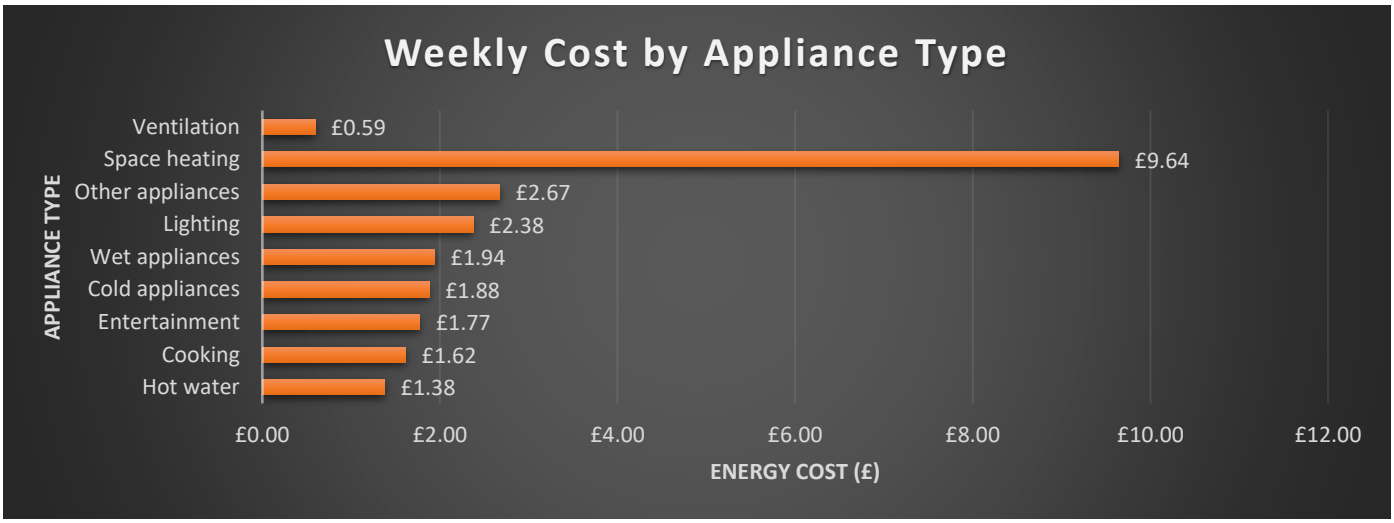


Figure 4: Categorical Breakdown of Weekly Energy Consumption.

As previously described, space heating consumed the most energy, followed by hot water and cooking. These appliance types all used gas, and cost less to run than the combined electrical consumption, as shown in figure 4.

Quarterly Total Energy Consumption Breakdown.

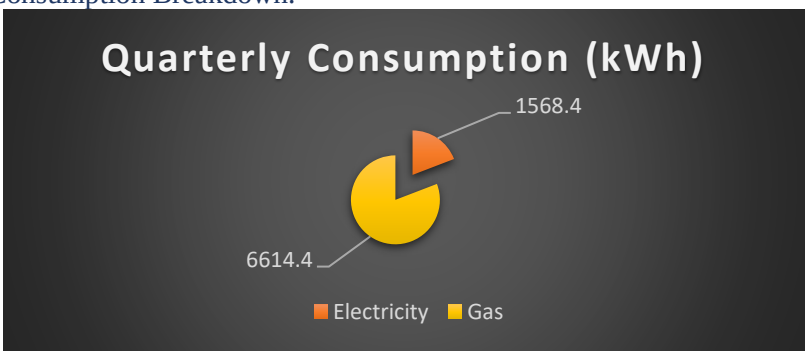


Figure 5: Quarterly Total Energy Consumption Breakdown by Fuel

Electricity accounted for 19.17% of the total quarterly consumption as represented by figure 5.

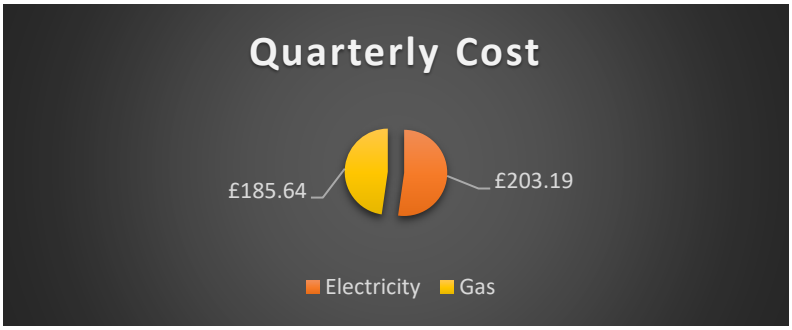


Figure 6: Quarterly Total Cost Breakdown by Fuel

Energy Source	Price (p/kWh)	Quarterly Consumption (kWh)	Quarterly Cost
Electricity	12.047	1568.4	£203.19
Gas	2.6762	6614.4	£185.64
	Total	**8182.8**	**£388.83**

Figure 7: Energy Data from Bill.

Figure 6 shows electricity had cost more per quarter than gas, due to the significant difference in price. Electricity was 4.8 times the gas price, as shown in figure 7, which made gas the economical energy source. The energy tariff was best tariff available at the time of writing, checked by using an energy supplier comparison website (Energy Comparison | Compare Gas & Electricity - Uswitch, 2021).

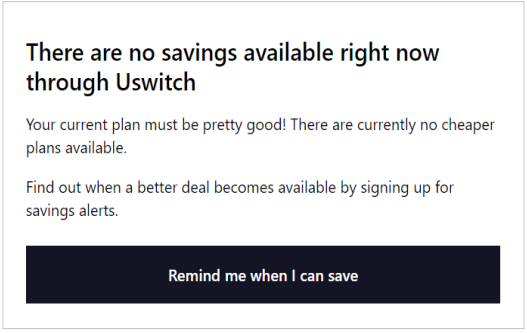


Figure 8: Snip of Results of Energy Supplier Comparison

Carbon Footprint Analysis

Energy Source	CO_2 (g/kWh)	Carbon Footprint (Tonnes)
Natural Gas	184 (Carbon tracker | Bulb, 2021)	1.217
Nuclear Electricity (Westinghouse Springfields, 2021)	20 (Carbon Footprint of Electricity Generation, 2011)	0.031
	Total	**1.248**

Figure 9: Carbon Footprint Breakdown.

Reducing carbon footprint

The carbon footprint can be reduced by using electric heaters rather than gas. Switching to electricity had cost more than using gas, but was a greener solution, with a total carbon footprint of 0.163 tonnes per quarter (Carbon Footprint of Electricity Generation, 2011); reducing the carbon footprint from gas by 89.13%.

Energy Loss Breakdown

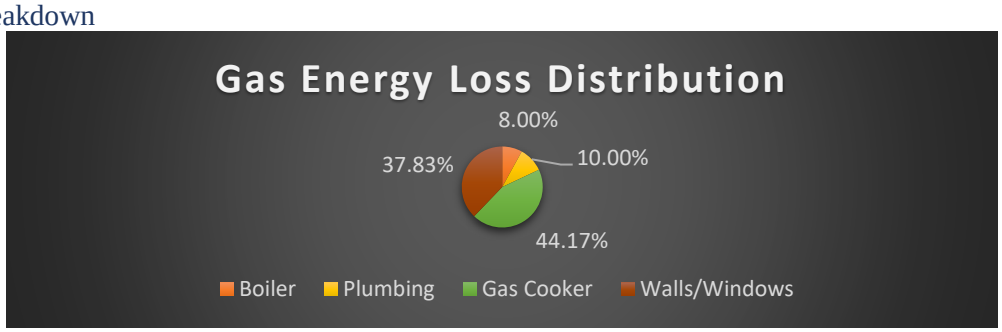


Figure 10: Distribution of Gas Energy Losses.

Losses	Quarterly Gas Loss by System (kWh)	Quarterly Gas Loss by Cost (£)
Boiler	74.223	£1.986
Plumbing	92.781	£2.483
Gas Cooker	409.814	£10.967
Walls/Windows	350.99	£9.393
Total	**927.81**	**£24.83**

Figure 11: Energy Losses from Gas

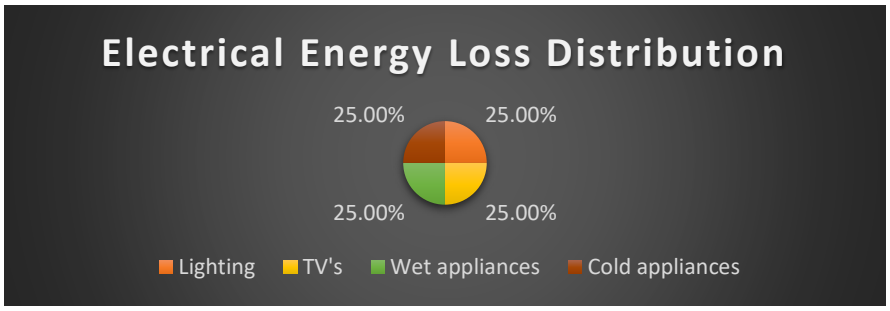


Figure 12: Distribution of electrical losses

Losses	Quarterly Electricity Loss by System (kWh)	Quarterly Electricity Loss by Cost (£)
Lighting	84.7508508	£10.89
TV's	84.7508508	£10.89
Wet appliances	84.7508508	£10.89
Cold appliances	84.7508508	£10.89
Total	**339.0034032**	**£43.55**

Figure 13: Energy Losses from Electricity

The distribution of the gas loss, shown in figure 10 was assumed by estimating the losses using statistics available online, while the electricity loss was assumed to be an even split between the 4 least efficient electrical appliances, shown in figure 12. As shown in the tables in figures 11 and 13, the energy loss equates to a significant amount of energy, which could be saved by improving the efficiency of the insulation and appliances of the home.

Efficiency Breakdown

Energy Source	Daily Energy in (From Bill)	Daily Energy out (From Data)	Efficiency
Electricity	17.427	13.391	76.84%
Gas	73.493	62.448	84.97%
Total / Average	**90.92**	**75.83891187**	**80.91%**

Figure 14: Home Energy Efficiency Calculation.

Figure 14 shows the overall efficiency of the home is 80.91%. The electrical energy losses were calculated to be greater than gas which may be due to transmission and transformation losses as heat and sound among others.

Conclusion

In conclusion, the following recommendations were made:

- Fit dimmers and PIR sensors to the lighting.
- Use less bulbs.
- Purchase efficient refrigeration.
- Close the door.
- Place the booster in the house.
- Switch to electric heating.
- Switch always on appliances off fully.
- Use the induction cooker more.
- Set the boiler output temperature to 60°C

The key economic issue found with reducing the carbon footprint was that electricity cost 4.8 times more than natural gas. The environmental issue with using natural gas was, the average life cycle of a gas field is 31 to 85 years (Darko, 2014),during which “1% to 9%” of its emissions are of methane, which was “86 times stronger at trapping heat than CO_2, over 20 years” (Environmental Impacts of Natural Gas, 2021). The social issue with using clean energy was that it had to be complemented by active changes to daily habits to reduce energy consumption thus load on the grid, reducing the homeowners carbon footprint. Thus, less energy would have to be produced, lowering the carbon footprint of the power plants, which was found to make a larger overall impact. Energy providers have to lower the unit cost of electricity using the joint effort of the communities reduced power consumption to build smaller power plants with a lower CO_2 output, encouraging the majority of households to switch to renewable energy sources as economically viable choices, At the time of writing, homeowners had saved on electricity bills by installing solar panels on their property to reduce the energy required from the grid. If the UK Government reached its planned “net zero” (Johnson and Sharma, 2020) goal for 2050, cheap electricity would have to be accessible.

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